

SOIYOD SIFAT MAHMUD

Electronics and Telecommunication Engineering
Chittagong University of Engineering and Technology (CUET)

01778871115 | soiyodsifat7@gmail.com | linkedin.com/in/soiyod-sifat-mahmud-548403407

PROFESSIONAL SUMMARY

Final-year Electronics and Telecommunication Engineering student with hands-on experience in software development, artificial intelligence, embedded systems, IoT, networking, and cybersecurity. Experienced in developing AI-powered applications, ESP32-based IoT systems, autonomous robotics solutions, and full-stack web applications. Strong programming background in Python, C/C++, JavaScript, React, FastAPI, and embedded development. Interested in AI/ML, software engineering, embedded systems, IoT, networking, and telecommunications.

EDUCATION

B.Sc. in Electronics and Telecommunication Engineering (ETE) **2026**
Chittagong University of Engineering and Technology (CUET), Bangladesh

TECHNICAL SKILLS

Programming: Python, C, C++, JavaScript, HTML, CSS

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Speech Emotion Recognition, OpenCV, YOLO, MobileNet SSD

Web Development: React, Vite, Tailwind CSS, FastAPI, Node.js, Express.js, REST API

Database: SQLite, MongoDB, ChromaDB, Google Sheets

Embedded/IoT: ESP32, Arduino, Raspberry Pi, MQTT, Sensors, Arduino IDE, Real-Time Monitoring

Networking/Security: Cisco Networking, Wireshark, Nmap, Packet Tracer, Cybersecurity Fundamentals

Tools: Git, GitHub, MATLAB, Flask, Tesseract OCR

PROJECTS

Mentora – AI-Powered PDF Learning Platform

- Developed an AI-powered learning platform that extracts and analyzes text, images, tables, diagrams, charts, mathematical equations, and handwritten content from PDF documents.
- Built the application using React, Vite, Tailwind CSS, FastAPI, PyMuPDF, Tesseract OCR, ChromaDB, SQLite, and LLM APIs.

Smart Temperature & Humidity Monitoring System

- Developed an ESP32-based IoT monitoring system using sensors, Arduino IDE, MQTT, database storage, and a custom web dashboard.
- Implemented real-time sensor monitoring, data visualization, analytics, and remote dashboard communication.

VisionDrive – Autonomous Monitoring Robot

- Developed an autonomous robot combining embedded systems, robotics, and edge AI for real-time object detection and counting.
- Used Arduino UNO for low-latency motor control and Raspberry Pi with MobileNet SSD and Flask MJPEG for real-time computer vision.

ProgressNest – Full-Stack Personal Growth Platform

- Built a full-stack platform for tracking personal goals, achievements, productivity, and collaborative learning activities.

Smart Home Safety & Security System

- Developed a real-time safety system integrating smoke and gas sensors, LDR, laser-based intrusion detection, person counting, Arduino control, and buzzer alerts.

INDUSTRIAL TRAINING

Industrial Attachment – Brain Station 23 **1 Week**

- Studied fintech applications in Bangladesh, including internet and mobile banking, BQR, digital payments, investment schemes, and financial products.
- Explored practical applications of artificial intelligence in financial technology and investment decision-making.
- Developed an AI-powered Investment Robo-Advisor Chatbot tailored to the Bangladeshi market to assist users in identifying suitable investment opportunities.

CERTIFICATIONS

- Engineering Design and Analyses with MATLAB** – Bangladesh Computer Council
- MATLAB Certified** – MATLAB
- Advent of Cyber** – TryHackMe
- Introduction to Cybersecurity** – Cisco Networking Academy
- Networking Basics** – Cisco Networking Academy